Reg. No. | | | | | | | | | | | [12IS8412]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

NETWORK SECURITY AND CRYPTOGRAPHY

Time: 3 Hours Maximum Marks:70

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. In a play fair cipher system, the encrypted message received read "IPDRETICETCAUTOLRS". What is the decrypted message if the key used was COMPUTERSCIENCE? List the rules used for it. **06**

   b. Alice wants to send Bob the message "CRYPTOGRAPHY IS COOL". **08**

   Alice choses the block size A=3, and chooses the key matrix. $K = \begin{bmatrix} 2 & 3 & 15 \\ 5 & 8 & 12 \\ 1 & 13 & 14 \end{bmatrix}$

   What is the cipher tent obtained if encryption algorithm used is hill cipher?

2. a. With a neat block diagram, explain single round of DES encryption. **06**

   b. Explain OFB encryption and decryption with a diagram. **08**

3. a. Describe RSA algorithm given various steps involved in it. Perform encryptions using RSA algorithm with the following: P=5, Q=7, E=7 and m=12, find D and cipher text. **08**

   b. Users A and B use the Diffie-Hellnar key exchange technique with a common prime q=71 and a primitive root $\alpha = 7$ **06**

   i. If user A has private key X_A =5 what is A's public key Y_A?

   ii. If user B has key private key XB=12, what if B's public key Y_B?

   iii. What is the shared secret key?

4. a. List and explain the requirements for message authentication. **06**

   b. With neat diagrams, explain how message authentication code can be used to provide message authentication and confidentiality. **08**

5. a. What is a digital signature? List the properties of digital signature and on basis of these properties discuss the requirements of digital signature. **08**

   b. Explain the working of digital signature algorithm (DSA). **06**

6. a. Discuss the difference between Kerberos 4 and Kerberos 5. **06**

   b. With a neat diagram, explain X.509 format. **08**

7. a. What is a PGP? List the reasons for its growth. 07

b. What is a S/MME? Explain the different header fields of MIME.

8. a. Why do we need IP level security? Explain four important applications of IP security 06

b. With a neat diagram, explain the ESP packet format.